Induction Heads and In-Context Grammar Learning in Mistral-7B

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Summary

Earlier, the per-layer analysis showed that in-context grammar learning becomes readable abruptly at **lens layer 19**, with a small step at lens layer 13. Here we asked which component produces this, testing **induction heads**: attention heads that copy what followed an earlier occurrence of the current word.

- **The strongest induction heads sit exactly where the learning appears:** in block 18, which writes lens layer 19, and in block 12, which writes lens layer 13.
- **Switching off 16 of the model's 1,024 heads cuts grammar learning sharply:** AGL from 0.75 to 0.37 of the bound, SRTT from 1.00 to 0.71. Switching off 16 random heads from the same blocks has no effect.
- **The general language ability is almost unchanged** (Wikipedia loss +0.8%), so this is not general damage.
- **The jump at layer 19 disappears** when the heads are switched off. About half of the learning still builds up gradually in later layers, so other components contribute too.
- **The heads carry rule knowledge, not only copying:** without them, the model no longer learns which AGL continuations are forbidden. A third test, adding noise at each layer, points the same way, more weakly.

Method

Induction heads (Olsson et al., 2022) implement the rule [A][B] … [A] → [B]: when the current word A appeared earlier, the head looks at the word that came right after it (B) and copies it into the prediction. They need no training on the task, only the context, so they are a natural candidate for in-context learning. Vaidya et al. (2023) found that in GPT-2 such attention appears sharply at one middle layer, much like our layer-19 jump.

Same setup as before: Mistral-7B (frozen, 8-bit), the same AGL and SRTT word streams (random part → grammar → random part), and the same learning score: (mean probability of the correct next word in the last 1,000 grammar words − the same in the first random part) ÷ (bound − that baseline). 50 runs per grammar.

Step 1: find the heads. Every one of the 1,024 heads (32 layers × 32) gets two scores:

- **Prefix matching** (does it *look* like an induction head?): on 25 random tokens repeated 4 times, the share of its attention that goes to the token right after the earlier copy of the current token. 0 = never, 1 = always.
- **Copying** (does it *act* like one?): this head's contribution alone, translated into words by the model's output layer. Where does the attended word rank among all ~32,000 words? +1 = it is the word the head pushes up most, 0 = chance, −1 = pushed down most.

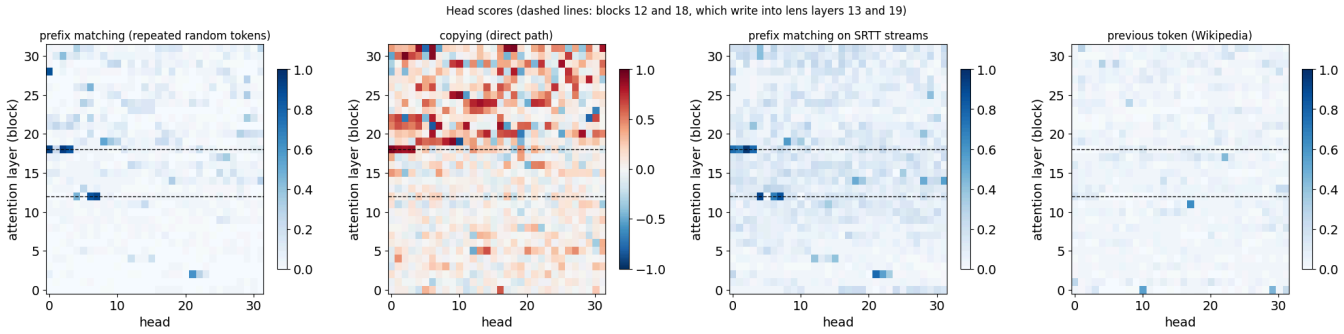
Heads are ranked by prefix × copying.

Step 2: switch them off (ablation). Each head writes 128 numbers into its layer's output. We replace only those 128 numbers with their average over Wikipedia text (mean ablation), so the head carries no information about the current input; all other heads are untouched.

Conditions: no ablation; the top 4, 8 and 16 heads; and **16 random heads with the same number per layer** as the top-16 set (2 draws). The random control is the key test: any network gets worse when parts are removed, so the effect only counts if the induction heads hurt far more than random heads from the same layers.

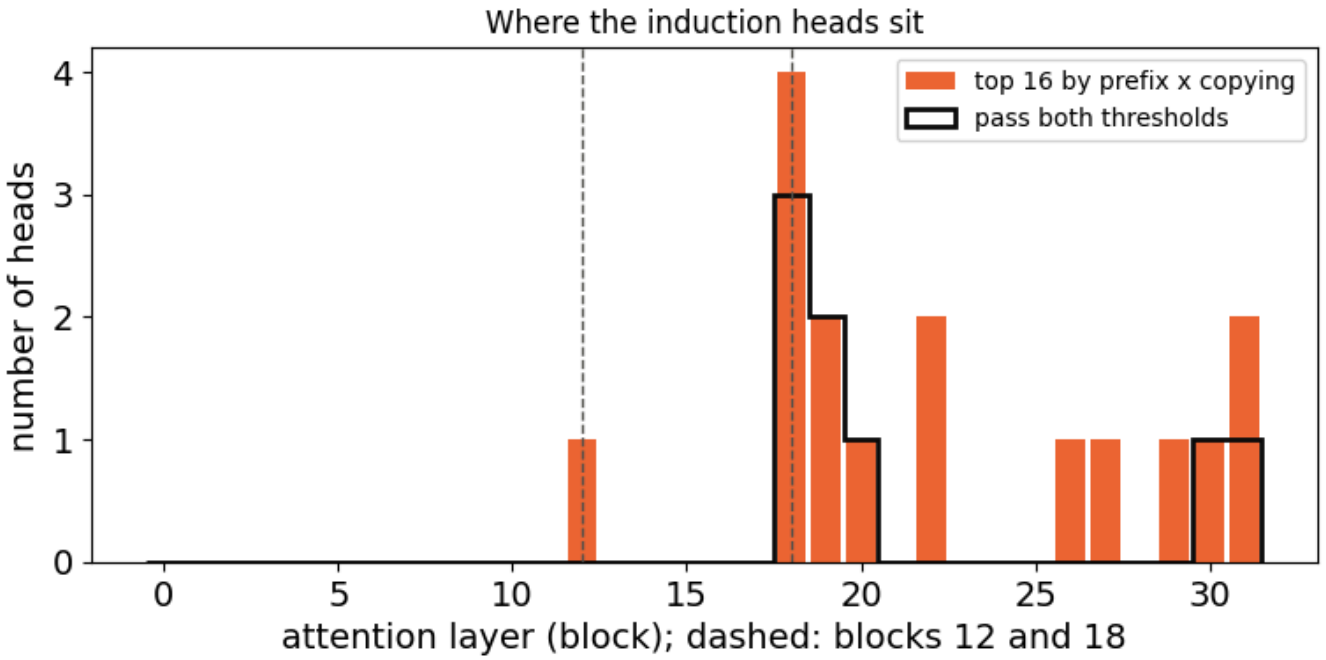
General-ability controls: next-word loss on held-out Wikipedia, and Olsson's in-context learning score (loss at token 500 minus loss at token 50).

Where the induction heads are



Rows = blocks (0 at the bottom), columns = heads; dashed lines = blocks 12 and 18. A note on numbering: **block 18 writes lens layer 19**, and block 12 writes lens layer 13 (lens layer 0 is the word embedding).

- **Two clusters of strong prefix matching (score 0.85–0.93):** block 18 (heads 0, 2, 3) and block 12 (heads 6, 7). These are exactly the blocks that produce the jump at lens layer 19 and the step at lens layer 13.
- **The block-18 heads also copy strongly (0.83–0.93),** so they are full induction heads. The block-12 heads attend in the induction pattern but copy weakly (~0.2): they likely pass information to later layers rather than writing the word directly.
- The same heads light up on our own SRTT streams (correlation with the random-token test r = 0.77), so they are active in our task.
- 8 heads pass both thresholds (prefix ⩾ 0.3, copying ⩾ 0.7). The top 4 are block 18 heads 0, 2, 3 and block 19 head 8.



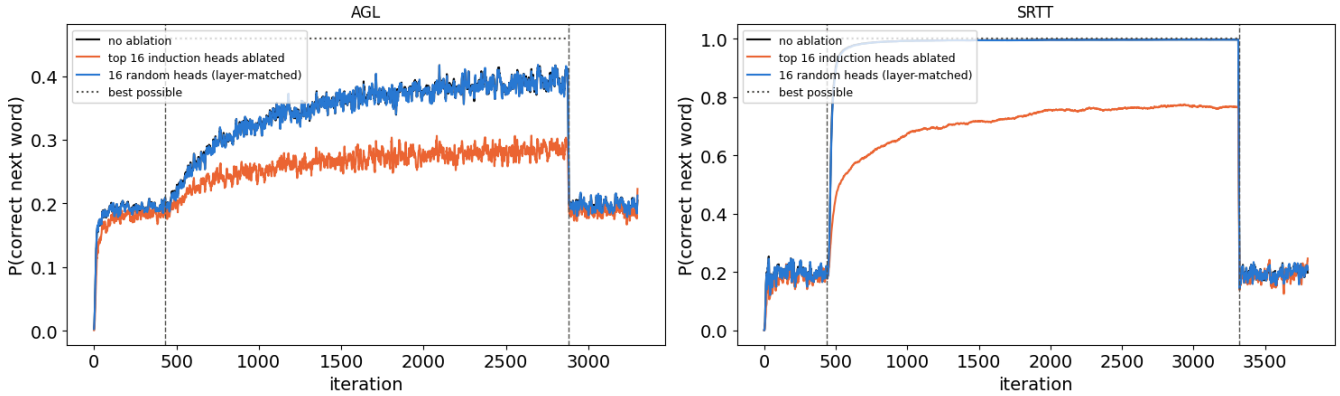
The top-16 set is concentrated in blocks 18–20, with single heads in block 12 and in the last blocks.

Ablation results

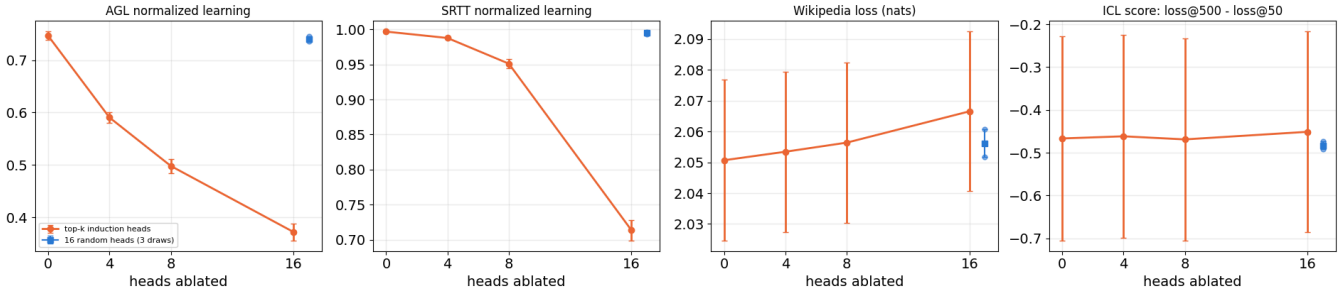
Condition	AGL learning	SRTT learning	Wikipedia loss	ICL score
No ablation	0.747	0.997	2.051	−0.467
Top 4 induction heads	0.590	0.988	2.053	−0.462
Top 8	0.498	0.951	2.056	−0.469

Condition	AGL learning	SRTT learning	Wikipedia loss	ICL score
Top 16	**0.372**	**0.713**	2.067	−0.451
16 random heads, same blocks (2 draws)	0.736–0.745	0.993–0.997	2.052–2.061	−0.474 to −0.491

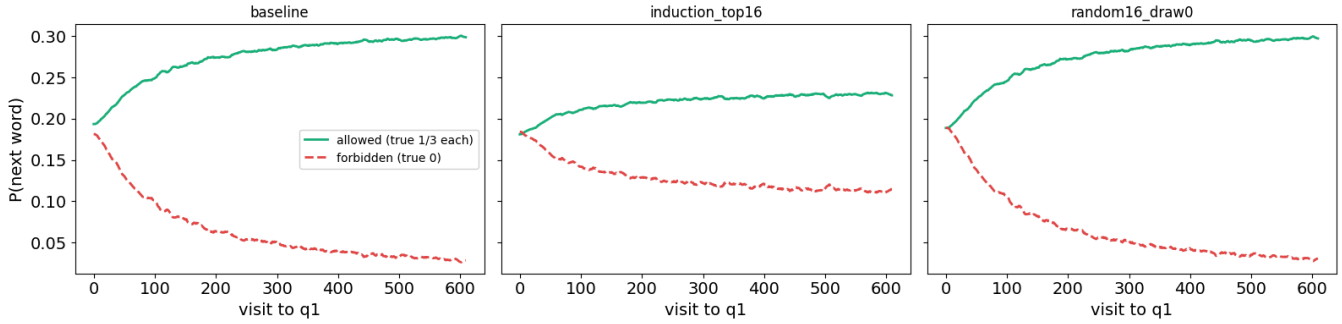
Learning: 1 = ideal learner, 0 = no learning (standard errors 0.008–0.017).



Learning curves. With the top 16 heads switched off (orange), AGL climbs to only ~0.28 instead of ~0.40, and SRTT levels off at ~0.77 instead of 1.0. Random heads (blue) lie exactly on the no-ablation curve (black, hidden underneath). Both curves still drop to chance after the grammar ends.

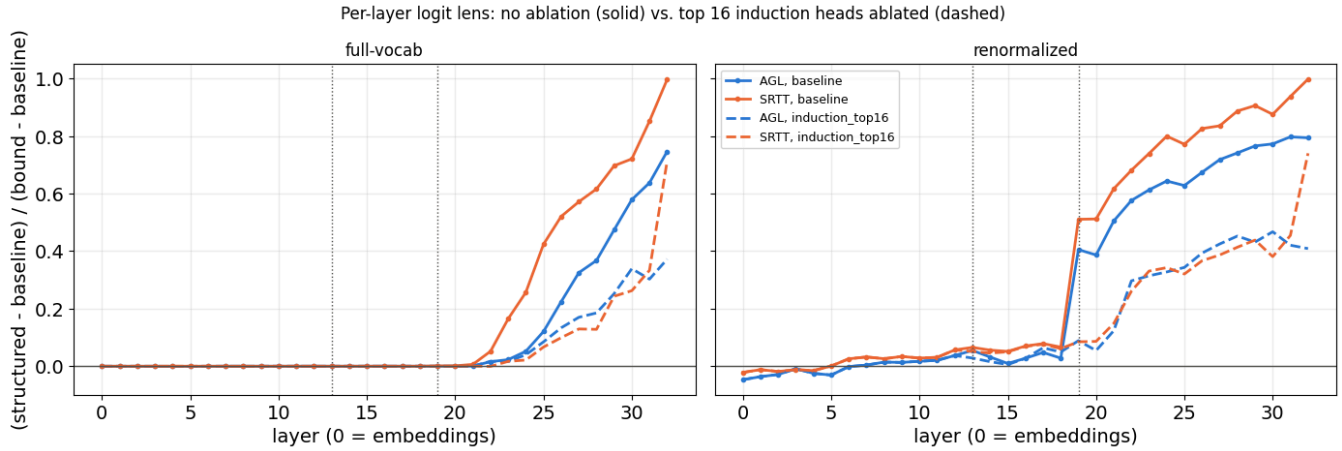


Dose-response. Learning falls steadily as more induction heads are removed: AGL already loses 21% with just 4 heads. Random heads (blue points) stay at baseline. Wikipedia loss rises only slightly (+0.8%, within the spread across texts), so the model is not broken in general. Olsson's ICL score on Wikipedia does not change, but its spread is very large, so it is too noisy to show an effect.



Rule knowledge (AGL state q1). At q1 three words are allowed and two are forbidden. Normally the forbidden ones fall to 0.03. With the induction heads switched off they only reach 0.11, and the allowed ones reach 0.23 instead of 0.30. The model loses much of its knowledge of *what cannot come next*, not just its ability to copy.

Per-layer view: the jump at layer 19 disappears



Solid = no ablation, dashed = top 16 induction heads switched off. Same per-layer score as in the earlier report.

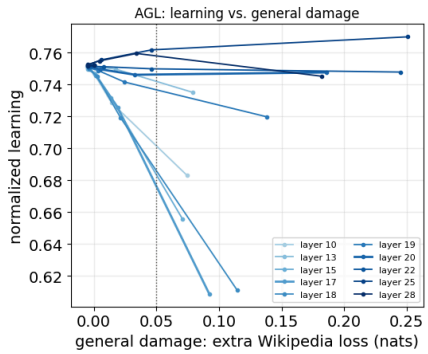
Lens layer	AGL, no ablation	AGL, ablated	SRTT, no ablation	SRTT, ablated
13	0.055	0.028	0.065	0.051
19	**0.405**	**0.089**	**0.511**	**0.085**
24	0.644	0.328	0.800	0.343
32	0.794	0.409	0.998	0.740

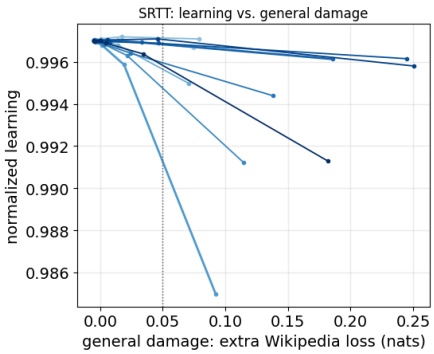
- **The abrupt jump at layer 19 is gone** for both grammars (0.41 → 0.09 and 0.51 → 0.09). The block-18 induction heads are what produce it.

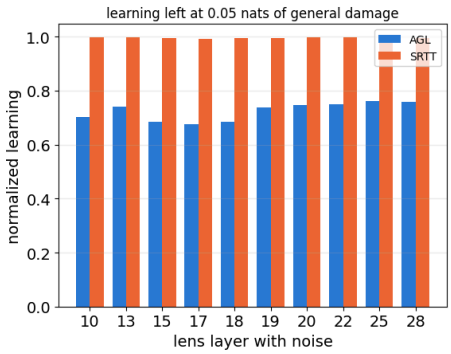
- **The step at layer 13 roughly halves for AGL** (only one of the two block-12 heads was in the ablated set).
- **Learning then builds up gradually from layer ~21**, reaching about half the normal level. Other components (weaker heads in later blocks, MLP layers) partly take over, but they do so slowly and incompletely.
- For SRTT, the last block recovers a lot (0.46 → 0.74 at layer 32), so the final block can partly compensate for the simple repeated sequence.

A third test: noise at each layer

Design. At one lens layer, every word's internal state gets a random push of relative size α (0.05 to 0.8 of that state's own size; BOS untouched). 10 layers × 5 noise sizes, 10 paired runs per grammar, plus a general-damage check on 25 Wikipedia texts. Layers are compared **at equal general damage** (extra Wikipedia loss), so "fragile" means fragile for the learning specifically, not for everything.







Lens layer with noise	10	13	15	17	18	**19**	20–28
AGL learning left at 0.05 nats of damage (no noise: 0.75)	0.70	0.74	0.68	0.68	0.69	**0.74**	0.75–0.76

- **Most fragile just before the jump.** Noise on the state the block-18 induction heads read (lens layers 15–18) costs about 10% of AGL learning; noise right after them (lens layer 19 onward) costs nothing.
- **Late layers dissociate learning from general ability.** Noise at layers 22–25 hurts Wikipedia prediction the most (+0.25 nats at α = 0.8), yet AGL learning does not drop.
- **A hint at the earlier cluster:** layer 10 is somewhat fragile (0.70), layer 13, right after the block-12 heads, is not (0.74).
- **SRTT is almost untouchable** (worst 0.985 vs. 0.997): too redundant for noise to break.

Caveats. The effects are small: even α = 0.8 removes only ~19% of AGL learning, because random noise mostly points in directions the model does not use. The ±0.016 error bars are the spread of the level across runs, not of the paired difference from baseline; the paired test, and error bars on the damage-matched values (which are interpolated), are still to be added. The noise also hits the current word and the earlier words at once, so it cannot separate the two. This is the weakest of the three lines of evidence, but it points the same way.

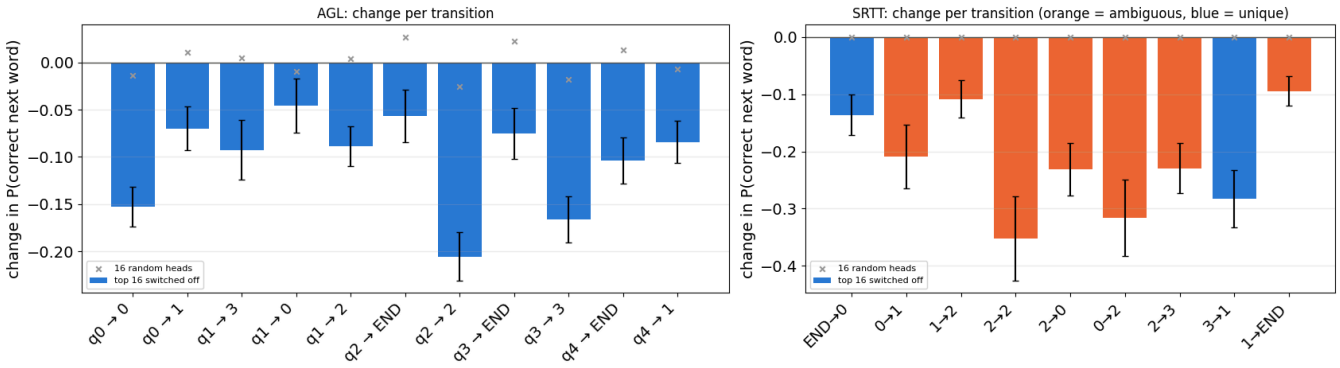
A fourth test: what each head does, per transition

The result: in SRTT the block-18 induction heads are interchangeable copies of one mechanism, and any one of them nearly suffices. In AGL they are partly divided, some pushing toward END and others toward continuing, and the learning needs their combination.

Design. Learning is split by transition (SRTT: 9 positions in the cycle, 7 ambiguous and 2 unique given the previous word; AGL: 11 state → word transitions). Two tests, 20 paired runs per grammar: (1) switch off the top 16 heads and see which transitions suffer; (2) **leave-one-in**: switch off all 16 except one, and measure what that single head restores.

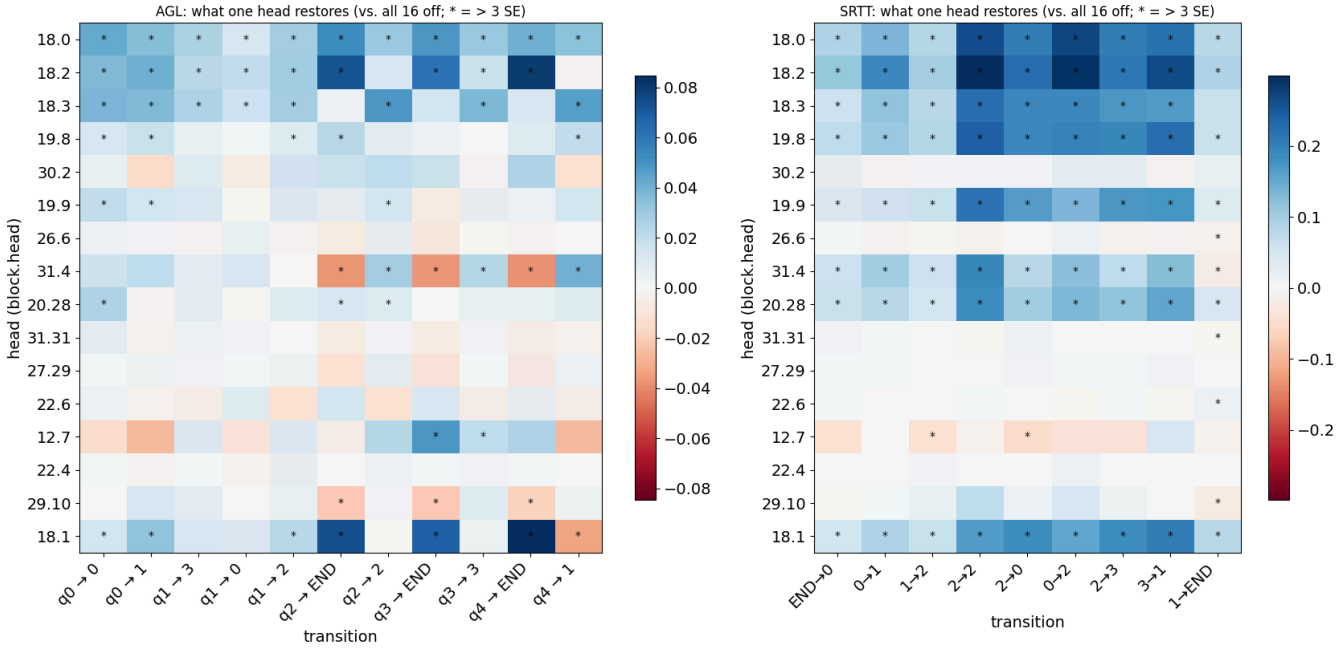
Which transitions suffer. All of them, in both grammars; 16 random heads change almost nothing.

- **SRTT shows no ambiguity effect:** ambiguous transitions drop 0.220, unique ones 0.210 (difference −0.011 ± 0.035). The heads are not used specifically where the previous word is not enough.
- **AGL:** hardest hit are q2→2 (−0.21), q3→3 (−0.17) and q0→0 (−0.15); END transitions drop least (−0.06 to −0.10).



What one head restores (leave-one-in).

	SRTT	AGL
Loss with all 16 off (mean)	0.216	0.104
Best single head	18.2 restores 0.198 (~92%)	18.2 restores 0.036 (~35%)
Other strong heads	18.0 ~80%, 19.8 ~71%, 18.3 ~65%, 18.1 ~61%	18.0, 18.1, 18.3: 0.027–0.035
Pattern across transitions	Even: no head prefers any transition	Split: 18.1 and 18.2 restore END (+0.06 to +0.085); 18.3 restores word continuations; 31.4 does the opposite of 18.1/18.2



- **SRTT is highly redundant.** One block-18 head brings back most of the learning. This explains why removing the top 4 heads barely hurt SRTT (0.988).
- **AGL needs several heads together,** matching the steady dose-response (the top 4 alone already cost 21%).
- **The AGL split is about which kind of word to output (END vs. continue), not about grammar states.** No head restores one state's transitions while leaving another's alone.
- **12.7 (block 12) restores almost nothing on its own,** consistent with it feeding block 18 rather than acting alone.

Caveats. Leave-one-in shows what one head can do inside a damaged network. The specialization index is meaningless for heads that restore ≈0. Significance stars (> 3 SE) are uncorrected over 198 cells. Next: turn back on pairs (18.2 + 18.3) and all four block-18 heads, to test whether the END and continue heads add up.

Conclusions

1. **Induction heads are a main mechanism of this in-context grammar learning.** Switching off 16 of 1,024 heads (1.6%) halves AGL learning and cuts SRTT learning by 28%, while 16 random heads from the same blocks do nothing and general language ability is nearly unchanged.
2. **They explain where learning appears in the network.** The strongest induction heads sit in block 18 and block 12, exactly the blocks behind the jump at lens layer 19 and the step at layer 13. Removing them erases the jump, and adding noise just before them hurts the learning more than noise after them.
3. **They support rule learning, not only literal copying.** In AGL, the next word is a random choice among allowed options, yet without these heads the model fails to learn which options are forbidden.
4. **They are not the whole story.** About half of the learning remains and builds up gradually in later layers, so other components contribute. This matches the warnings in both papers: in large models with MLPs, other parts can partly take over.
5. **Against the initial guess, AGL depends on these heads more than SRTT.** The repeated sequence keeps 71% of its learning and recovers at the last layer, while AGL keeps only half.
6. **The two grammars use the heads differently.** In SRTT any single block-18 head restores 60–90% of the loss, so the heads are redundant copies. In AGL a single head restores at most ~35%, and the block-18 heads divide between pushing toward END and toward continuing.

Limitations and next steps

- **Which cluster does what?** We ablated the top-k ranking, which is mostly block 18–20 heads plus one block-12 head. Next: switch off the block-12 cluster alone and the block-18 cluster alone, each with a random control from the same block, to test directly whether each one creates its own onset.

- **What carries the remaining half?** Next: ablate more heads (k = 32, 64) chosen by the corrected ranking, and check whether the gradual rise after layer 21 also disappears.
- **How do blocks 12 and 18 interact?** The block-12 heads look like induction heads but copy weakly; they may feed the block-18 heads. Path patching (switching off only the connection between them) would test this.
- **Other models:** repeat on Llama-3.1-8B to see whether the jump and its heads sit at the same relative depth.
- **Statistics:** 50 runs per grammar; all SRTT runs share one sequence, so SRTT error bars are narrower than for independent runs. The ICL score on Wikipedia is too noisy to be informative.
- **8-bit model** (memory limits), as in all earlier analyses.

Note on the first run. A first run had a bug in the copying score (it compared against all 32,000 words, so every head scored ~0). The ranking was therefore mostly noise, and the 32 ablated heads were largely weak ones; the block-12 cluster was not included at all. Even so, ablating them cut AGL learning from 0.75 to 0.41 (SRTT 1.00 → 0.87) while 32 random heads did nothing. The corrected run, shown above, gets a larger effect with half as many heads, which is consistent with the effect coming from the true induction heads.